

Predictive Modeling of YouTube Video Success: A Machine Learning Perspective

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Abstract—This study presents a systematic methodology for developing a machine learning pipeline to predict YouTube video success based on various features. The process encompasses data preprocessing, feature engineering, and model optimization. Key steps include handling missing values, encoding categorical variables, scaling features using Min-Max Scaling, and performing outlier removal. Target variable binning and feature selection were performed to enhance model performance and interpretability. Several classifiers, including Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost, were evaluated using hyperparameter tuning and cross-validation. Evaluation metrics such as accuracy, MAE, RMSE, R^2 , precision, recall, and F1-score were used to identify the best-performing model. The approach ensures robust and reliable predictions, with results demonstrating the efficacy of the methodology in handling diverse datasets. The novelty of this work lies in its application to Bangla YouTube videos, a relatively unexplored domain, and the use of a robust methodology incorporating diverse machine learning techniques to optimize predictions for this unique dataset. The Random Forest and XGBoost classifiers emerged as the most effective, with the tuned Random Forest achieving the highest performance metrics (86% accuracy), emphasizing the significance of hyperparameter tuning and ensemble approaches for complex datasets.

I. INTRODUCTION

With the exponential growth of digital platforms, YouTube has emerged as a dominant force in the dissemination of video content. Hosting billions of videos across diverse categories, it caters to creators, viewers, and advertisers alike. For content creators, understanding what makes a video successful is of paramount importance. Video success is often measured by metrics such as view counts, likes, comments, and shares. Predicting these metrics can provide valuable insights into audience behavior and guide creators in optimizing their content strategy.

Despite extensive research on YouTube video success prediction, existing studies have largely focused on global datasets or specific regional contexts, neglecting the unique dynamics of Bangla-language content. This study addresses the gap by analyzing Bangla YouTube videos, which are gaining significant traction due to the rising popularity of regional content on digital platforms. The primary problem tackled in this research is: *Can machine learning models trained on metadata and engagement metrics reliably predict the popularity of Bangla YouTube videos?*

The importance of this problem lies in its potential to empower Bangla content creators with actionable insights.

By identifying critical factors influencing video success, this research aids creators in optimizing their content for greater reach.

To address this challenge, we propose a robust pipeline that combines data preprocessing, feature engineering, and machine learning algorithms. Unlike previous approaches, our work leverages a novel dataset curated using the YouTube Data API V3, comprising over 16,000 Bangla YouTube videos across 10 categories. The dataset captures both metadata (e.g., title length, description length, category ID) and engagement metrics (e.g., likes, comments, views). This is, to the best of our knowledge, the first study applying machine learning to Bangla YouTube video prediction.

Key Contributions

The primary contributions of this research are as follows:

- 1) **Novel Dataset:** Development of a comprehensive dataset of Bangla YouTube videos, enabling regional video analytics.
- 2) **Critical Feature Identification:** Analysis of features that significantly influence video performance, such as title length, like count, and category ID.
- 3) **Model Development:** Implementation and evaluation of multiple machine learning models, including Random Forest, XGBoost, and K-Nearest Neighbors, optimized using hyperparameter tuning.
- 4) **Predictive Framework:** Creation of a scalable machine learning pipeline for regional content prediction, addressing the unique challenges of Bangla videos.

We hypothesize that: *Machine learning models trained on metadata and engagement metrics can effectively predict the popularity of Bangla YouTube videos, achieving high accuracy and reliability.*

The rest of this paper is structured as follows: the next section II reviews related literature, highlighting gaps in existing works. Section III details the methodology, including dataset preparation, feature engineering, and model evaluation. Section IV presents the results obtained from various classifiers and hyperparameter tuning. Section V discusses the findings and their implications, while Section VI discusses about the Limitation of the work and Section VII concludes the study with key insights and future research directions.

II. LITERATURE REVIEW

Due to the platform's rapid growth and success because of the monetary incentive and its role as vast entertainment media, there has been a natural increase of curiosity in people and researchers. So far, there has been considerable amount of work done to predict the success in YouTube videos to optimize the content strategies to adapt for creators. This section briefly explores significant work that has been done in this area.

A. Related work on YouTube video popularity prediction using machine learning

Due to the speedy growth of the platform along with the machine Learning techniques, there has been many studies performed to predict video popularity on private and public datasets. One of the significant studies performed on this sector is "Optimizing Prediction of YouTube Video Popularity Using XGBoost". [1] To predict the highest viewership on YouTube videos, this study performed XGBoost model, considering features selection, fusion, min-max normalization and some precision parameters such as gamma, eta, learning rate etc. The dataset was collected using YouTube API and total of 15 features were used to train the model. This model obtained very promising results giving 86% accuracy, 64% precision, 67% recall and 65% F1 score. Additionally, another study "YouTube Videos Prediction: Will this video be popular?" has been performed to analyze video content optimization. [2] This study focused on multi-class algorithms like stochastic gradient descent, multilayer perceptron neuron network, decision trees, random forest, extreme gradient boosting, gradient boosting method are used. The predicted output classes are non-popular, overwhelming praises, overwhelming bad views, neutral videos. The datasets used in this study were collected from Trending YouTube Video Statistics dataset from Kaggle and selected the data of trending videos of YouTube from the US and Canada. Total of 29089 videos information were used to build the dataset. Furthermore, there has been another study "Predictive analysis of YouTube trending videos using Machine Learning" to analyze factors that influences the likelihood of a video being viral. The dataset for this study was collected from Kaggle, [3] it explores the attributes of like Views, likes, dislike and comment ability to impact a video to be in trend. This study explores various classification model for instance Random Forest, SVM, Decision Tree, Logistic Regression, and Gaussian Naïve Bayes. SVM and random forest model showed the most promising result. It also applies CRISP-DM methodology for data mining.

B. Related work on YouTube video popularity prediction using sentimental analysis and deep learning

Using machine learning to capture the full dynamics and complexities of YouTube video success may not be fully possible. Several studies have been done by integrating using sentimental analysis and deep learning to provide meaningful and more deeper insights. A study "YOUTUBE TRENDING VIDEOS PREDICTION & ANALYSIS" has been done using

sentiment analysis to discover the strategies behind increasing popularity in videos. It uses user comments and analyses it to discover trending videos that the users will want to see and videos related to user video types. [4] The study uses support vector algorithm (SVM) identify which videos will become popular on the platform. Additionally, another study "YouTube Video Popularity Analysis & Prediction Using Deep Learning" aims predict YouTube video success using deep learning (DL). It was conducted during the Covid-19 pandemic which led to the rise in YouTube usage, which provided a unique opportunity to analyze the factors. The dataset that was used in this study is quite diverse as it crosses over various categories and viewership.

See Table I to understand the differences in Methodology and Limitations among these related works.

III. METHODOLOGY

The methodology begins with data cleaning and preprocessing, including the handling of missing values, removal of duplicates, and encoding of categorical variables. Target variable binning ensures a balanced representation of view count categories, while transformations like logarithmic and Yeo-Johnson address skewness in the data. Feature scaling using Min-Max Scaling and feature selection techniques further enhance the model's performance by reducing noise and dimensionality.

To ensure robustness, multiple machine learning algorithms were evaluated, including Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost. Hyperparameter tuning was conducted using RandomizedSearchCV, and Stratified K-Fold Cross-Validation was employed to validate the models' generalizability. Evaluation metrics such as accuracy, MAE, RMSE, and R^2 provided a comprehensive understanding of each model's predictive capabilities.

A. Dataset Description

The dataset consists of metadata and performance statistics of YouTube videos, including attributes such as title length, description length, like count, comment count, duration, and categorical variables such as category ID. The target variable, *View Count*, represents the total number of views per video.

B. Data Pre-processing

- 1) **Handling Missing and Duplicate Values:** Missing values were identified and removed to ensure data integrity. Duplicate rows were also eliminated.
- 2) **Column Selection:** Irrelevant columns such as *Title*, *Tags*, *Description*, *Channel Name*, and *Channel ID* were dropped based on domain knowledge.

C. Feature Transformation

To address outliers and improve feature scaling, the following transformations were applied:

- **Logarithmic Transformation:** Features with highly skewed distributions, such as *Like Count*, *Total Videos* and *Total Views*, were log-transformed using:

$$x_{\log} = \log(x + 1) \quad (1)$$

TABLE I: Comparison of Related Works

Reference	Study	Methodology	Dataset	Strength	Limitations
[1]	Optimizing Prediction of YouTube Video Popularity Using XGBoost.	XGBoost, parameter tuning	Self-curated	High performance with 86% accuracy	Focus on features diversity lacking
[2]	YouTube Videos Prediction: Will this video be popular?.	Multi-class algorithms (SGD, RF, XGB)	Kaggle (US & Canada)	Feature selection effective	Limited dataset
[3]	Predictive analysis of YouTube trending videos using Machine Learning.	Random Forest, SVM, Decision Tree, Logistic Regression, and Gaussian Naïve Bayes	Kaggle	Lifecycle prediction of trending videos	Accuracy low (62.53%)
[4]	Youtube trending videos prediction & analysis.	Support Vector Algorithm (SVM)	Gaming videos dataset	Insights on user preference	Dataset not diverse (Focuses on gaming content only))
[5]	YouTube Video Popularity Analysis & Prediction Using Deep Learning .	Deep Learning	Self-curated (Diverse and Large)	Robust and scalable	Less practical applicability focus

- **Yeo-Johnson Transformation:** For certain features with left skewed, such as *Title Length*, the Yeo-Johnson method was applied for optimal normality, represented by:

$$f(x; \lambda) = \begin{cases} \frac{(x+1)^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \log(x+1) & \lambda = 0 \end{cases} \quad (2)$$

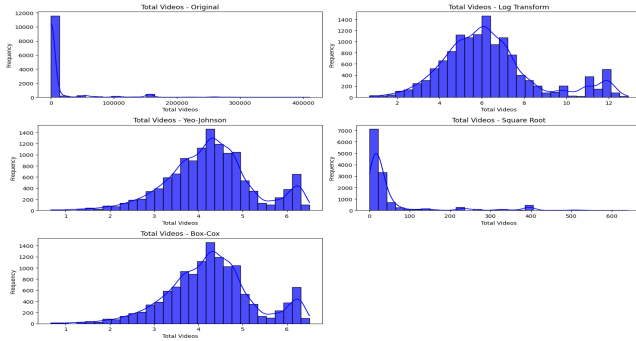


Fig. 1: Transformation Comparison between Original, Log, Sqrt, Yeo-Johnson and Box-Cox

D. Categorical Data Preprocessing

Categorical features were transformed into numerical representations:

- **Mapping Unique Values:** Features such as *Published Week Day*, *Publish Time*, and *Definition* were encoded into numerical values.
- **Category Analysis:** Sparse categories were identified, and *Category ID 23*, containing only one instance, was dropped for better generalization.

E. Exploratory Data Analysis

- **Outlier Removal:** Extreme values were removed by eliminating the lower 0.2% and upper 0.2% data points in numerical characteristics.

$$IQR = Q_3 - Q_1 \quad (3)$$

F. Target Variable Binning

The target variable (*Total Views*) was binned into five categories based on predefined thresholds:

- 1) 0 – 1k
- 2) 1k – 50k
- 3) 50k – 500k
- 4) 500k – 3M
- 5) > 3M

These thresholds ensured an unbiased distribution across the target variable.

G. Train-Test Split

The dataset was split into training and testing sets with an 80 : 20 ratio. Stratified sampling ensured consistent target variable distributions across splits.

```

Train set distribution:
View Count Category
1k-50k      0.243101
0-1k       0.214656
50k-500k   0.206539
500k-3M    0.185592
>3M        0.150112
Name: proportion, dtype: float64

Test set distribution:
View Count Category
1k-50k      0.243277
0-1k       0.214529
50k-500k   0.206491
500k-3M    0.185781
>3M        0.149923
Name: proportion, dtype: float64

```

Fig. 2: Stratified split

H. Feature Scaling

Min-Max Scaling was employed to bring all features to a uniform range $[0, 1]$, using the formula:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (4)$$

I. Feature Selection

Recursive Feature Elimination (RFE) was used to select the most important features, optimizing for model performance while minimizing dimensionality. Features that have less importance (< 0.03), were removed.

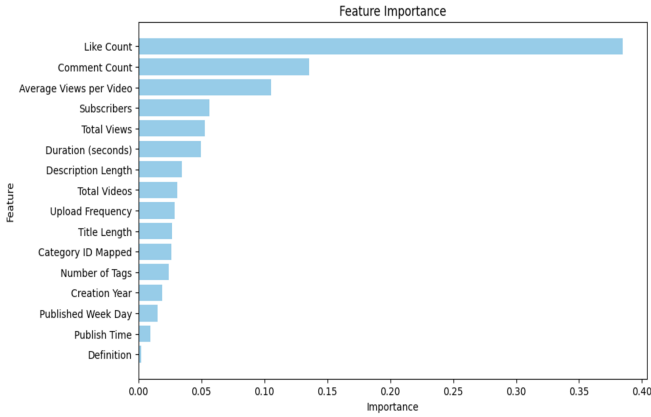


Fig. 3: Feature Importance

J. Correlation matrix

If two features were highly correlated with each other, the less important feature was dropped.

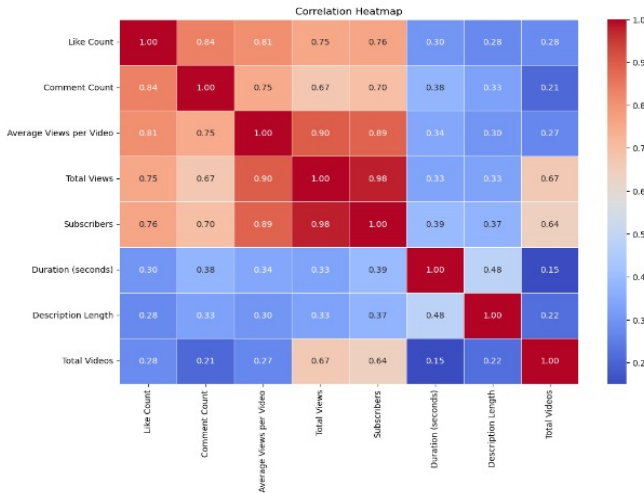


Fig. 4: Correlation Matrix before dropping higher correlated columns

K. Model Selection and Hyperparameter Tuning

Several machine learning classifiers were evaluated with hyperparameter tuning using RandomizedSearchCV. The configurations for each classifier are detailed in Table II.

TABLE II: Classifier Configurations

Classifier	Hyperparameter Configurations(Tuned)
Decision Tree (DT)	splitter='best', min_samples_split= 15, min_samples_leaf= 2, max_depth= 10, criterion= 'gini'
Random Forest (RF)	n_estimators= 300, min_samples_split=10, min_samples_leaf= 1, max_depth= 50, criterion= 'entropy', bootstrap= False
K-Nearest Neighbors (KNN)	weights= 'distance', p= 2, n_neighbors= 21, metric= 'manhattan'
XGBoost (XGB)	subsample=0.8, n_estimators=200, max_depth=6, learning_rate=0.1, colsample_bytree=1.0, random_state=42

L. Model Training and Evaluation

- 1) **Model Selection:** Multiple classifiers, including Random Forest, Decision Tree, K-Nearest Neighbors, and XGBoost, were trained and evaluated using Stratified K-Fold Cross-Validation to mitigate data imbalance.
- 2) **Hyperparameter Tuning:** RandomizedSearchCV was used to optimize hyperparameters for the K-Nearest Neighbors, Random Forest, Decision Tree, and XGBoost models. For example, the parameters for XGBoost included:
 - Number of estimators ($n_{\text{estimators}}$)
 - Learning rate (learning_rate)
 - Maximum depth (max_depth)
- 3) **Evaluation Metrics:** The performance of each model was assessed using the following metrics:

- Training and Testing Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

- Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (6)$$

- Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

- R^2 Score

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

- Precision, Recall, and F1-Score

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (10)$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

M. Flow Diagram

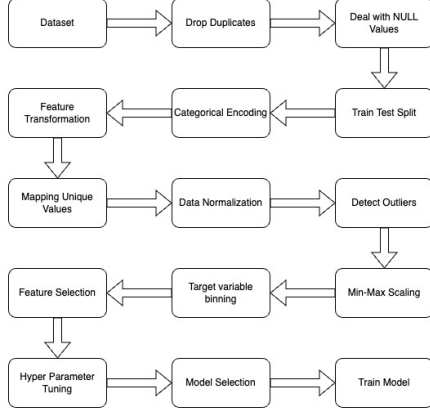


Fig. 5: Flow diagram of the methodology.

N. Explainable AI Using LIME

To enhance the interpretability of the machine learning model, we applied Local Interpretable Model-Agnostic Explanations (LIME) to our best-performing model, the tuned Random Forest (RF1). LIME enables us to understand the contribution of individual features to the model's predictions both globally and locally.

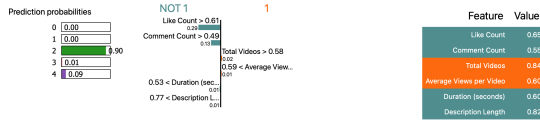


Fig. 6: Global explanation using LIME

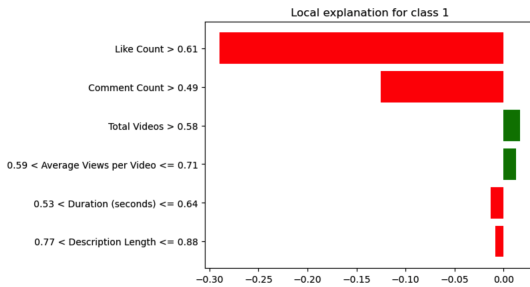


Fig. 7: Local explanation using LIME

This approach allowed us to identify the most critical features driving the model's predictions and ensured that the model's decisions were aligned with domain knowledge. By integrating LIME into the methodology, we addressed the challenge of interpretability, a common limitation in complex machine learning models like Random Forest.

O. Summary of Methodology

The outlined steps ensure a robust and reproducible pipeline for predicting YouTube video views, addressing data challenges such as outliers, skewness, and high dimensionality.

IV. RESULTS

The performance of various machine learning classifiers was evaluated in this study to predict YouTube video popularity effectively. The evaluation considered multiple metrics, including train accuracy, test accuracy, mean absolute error (MAE), root mean squared error (RMSE), R^2 score, and overall accuracy. Tables III and IV summarize the results, providing a detailed comparison of each algorithm's performance under different configurations.

A. Model Accuracy Analysis (Table III)

Table III demonstrates the train and test accuracy for different classifiers:

- **Decision Tree (DT):** The Gini and Entropy variants of the Decision Tree algorithm both achieved perfect train accuracy (1.000), indicating potential overfitting, as their test accuracies dropped to 0.794 and 0.793, respectively. After tuning, the Entropy-based Decision Tree exhibited better generalization, achieving a test accuracy of 0.810.
- **Random Forest (RF):** Default configurations of the Random Forest algorithm showed high accuracy on both train and test datasets, with the test accuracy reaching 0.852. Modifying the number of trees to 80 with square-root splitting criteria further improved the test accuracy slightly to 0.856. After hyperparameter tuning, Random Forest achieved the best test accuracy among all classifiers at 0.858, indicating its robustness.
- **K-Nearest Neighbors (KNN):** The default configuration of KNN ($k = 3$) showed moderate performance with a test accuracy of 0.797. After tuning, the algorithm achieved a test accuracy of 0.823, demonstrating a significant improvement in its predictive capabilities.
- **XGBoost (XGB):** The default configuration of XGBoost achieved 0.850 test accuracy. With hyperparameter tuning, the performance slightly improved to 0.851 on the test set, showcasing its effectiveness in handling complex data structures.

TABLE III: Model Accuracy Results

Classifier	Train Acc.	Test Acc.
Decision Tree (Gini)	1.000	0.794
Decision Tree (Entropy)	1.000	0.793
Decision Tree (Entropy, Tuned)	0.878	0.810
Random Forest (Default)	1.000	0.852
Random Forest (80 Trees, sqrt)	1.000	0.854
Random Forest (Tuned)	0.998	0.856
K-Nearest Neighbors (Default)	0.893	0.797
K-Nearest Neighbors (Tuned)	1.000	0.823
XGBoost (Default)	0.984	0.850
XGBoost (Tuned)	0.969	0.851

B. Performance Metrics Analysis (Table IV)

Table IV provides an in-depth comparison of various performance metrics across the algorithms:

- **Mean Absolute Error (MAE):** The tuned XGBoost (XGB1) and Random Forest (RF2) achieved the lowest MAE (0.215 and 0.210, respectively), indicating that these models make smaller average errors in predictions compared to other classifiers.
- **Root Mean Squared Error (RMSE):** Random Forest (RF2) exhibited the lowest RMSE value (0.587), demonstrating its effectiveness in minimizing larger prediction errors. XGBoost (XGB1) followed closely with an RMSE of 0.590.
- **R^2 Score:** The tuned Random Forest (RF2) achieved the highest R^2 score of 0.821, indicating its superior ability to explain the variance in the data. XGBoost (XGB1) also performed comparably with an R^2 score of 0.819.
- **Overall Accuracy:** Random Forest (RF2) emerged as the most accurate model with 86% accuracy, followed closely by XGBoost (XGB1) at 85%.

TABLE IV: Performance Metrics of Different Machine Learning Classifiers

Classifiers	MAE	RMSE	R^2 Score	Accuracy
DT (Gini)	0.302	0.706	0.741	0.80
DT1 (Entropy)	0.304	0.704	0.742	0.79
DT2 (Entropy, Tuned)	0.280	0.678	0.761	0.81
RF (Default)	0.218	0.599	0.813	0.85
RF1 (80 Trees, sqrt)	0.208	0.586	0.821	0.85
RF2 (Tuned)	0.210	0.587	0.821	0.86
KNN (k=3)	0.303	0.709	0.738	0.80
KNN1 (Tuned)	0.260	0.635	0.778	0.82
XGB (Default)	0.216	0.591	0.818	0.85
XGB1 (Tuned)	0.215	0.590	0.819	0.85

The results highlight the Random Forest and XGBoost algorithms as the most effective classifiers for predicting YouTube video popularity. Among these, the tuned Random Forest model demonstrated the best overall performance across all evaluation metrics, achieving the highest test accuracy (0.857), lowest RMSE (0.587), and highest R^2 score (0.821). These results underscore the importance of hyperparameter tuning and the utility of ensemble methods in handling complex datasets like YouTube video statistics.

V. DISCUSSION

The aim of this study was to predict the popularity of Bangla YouTube video content using machine learning algorithms. While implementing this work, we have found several factors contributing to the widespread of video content. Among them we've found that Like count, Comment count, Average videos per video and subscribers are the highest indicator to determine the popularity of a video. We've used feature selection and removing irrelevant features to further the power of predictability of the classifier. We've applied total of 4 algorithms, K-nearest neighbor, Decision Tree, Random Forest and XGBoost. We found the best prediction result using Random Forest

algorithm with 86 %. Random forest produces the best result because it has the strength of various tree and uses ensemble approach. Our study is distinguishable and stands out from the highlighted work at the table with high accuracy and unique dataset.

TABLE V: Comparison of this study and other related works

Ref.	Study	Dataset	Used Algorithms	Best Model	Evaluated Metrics
[1]	Optimizing Prediction of YouTube Video Popularity Using XGBoost.	Self-curated using YouTube API	Decision Tree, XGBoost, Tuned XGBoost	Tuned XG-Boost	88% accuracy, 84% precision, 63% recall
[2]	YouTube Videos Prediction: Will this video be popular?.	Trending YouTube Video Statistics dataset from Kaggle (US and Canada)	Decision Tree, Random Forest, XGBoost, MLPC, SGDC	XGBoost	73.6% F1-Score
[3]	Predictive analysis of YouTube trending videos using Machine Learning.	Unsupervised data from kaggle	Random Forest, SVM (Support Vector Machine), Decision Tree, Logistic Regression, Gaussian Naïve Bayes and Linear Regression	Random Forest classifier	62.53% accuracy
[4]	Youtube trending videos prediction & analysis.	YouTube comments and engagement metrics.	KNN (K-Nearest Neighbors), SVM (Support Vector Machine)	SVM (Support Vector Machine)	Accuracy not stated explicitly but model performed steadily
[5]	YouTube Video Popularity Analysis & Prediction Using Deep Learning.	Self-curated using YouTube API	Deep Learning Framework	Deep Learning Framework	Overall accuracy not explicitly reported but 6 out of the first 10 videos were correctly predicted
This Work	Predictive Modeling of YouTube Video Success: A Machine Learning Perspective	Self-curated using YouTube API	XGBoost, KNN, Decision Tree and Random Forest	Random Forest	86% accuracy

VI. LIMITATIONS OF THE WORK

Our model shows promising results on the test dataset however despite the novel approach of this project, there are some limitations and setbacks within the study that could

hinder the model's accuracy and applicability in real life. Some of these key challenges are-

- 1) **Limited Features:** While our study includes some essential attributes for prediction such as Likes, Views, Comments etc., it doesn't take full account of some other potential drivers such as viewer sentiment, collaborations, social media activity, engagement through shares, viewer retention time etc.
- 2) **API Constrains:** While API can be very powerful and comprehensive tool, it does have rate limits which can provide incomplete or outdated data that can affect the accuracy of the dataset.
- 3) **Real-Time Trends:** Since we have retrieved the data using API, the model is essentially using historical data that does not account for or accurately reflect the rapid change in trends and viral videos. This makes our model lack in terms to predict viral videos accurately.

VII. CONCLUSION & FUTURE WORK

In conclusion, this study bridges the gap between raw metadata and actionable insights, offering a scalable framework for predicting video performance on YouTube. By combining rigorous data preprocessing with advanced machine learning techniques, it provides a practical solution to a pressing challenge in the digital content ecosystem.

While this study focused on Bangla YouTube videos, future research can expand the scope to include other regional languages and diverse content categories. Additionally, integrating advanced deep learning architectures, such as transformer models, may further enhance prediction accuracy. Exploring user sentiment analysis and video thumbnail effectiveness are promising directions to enrich the predictive framework. Finally, deploying this framework at scale with real-time prediction capabilities could provide further utility for creators and marketers.

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